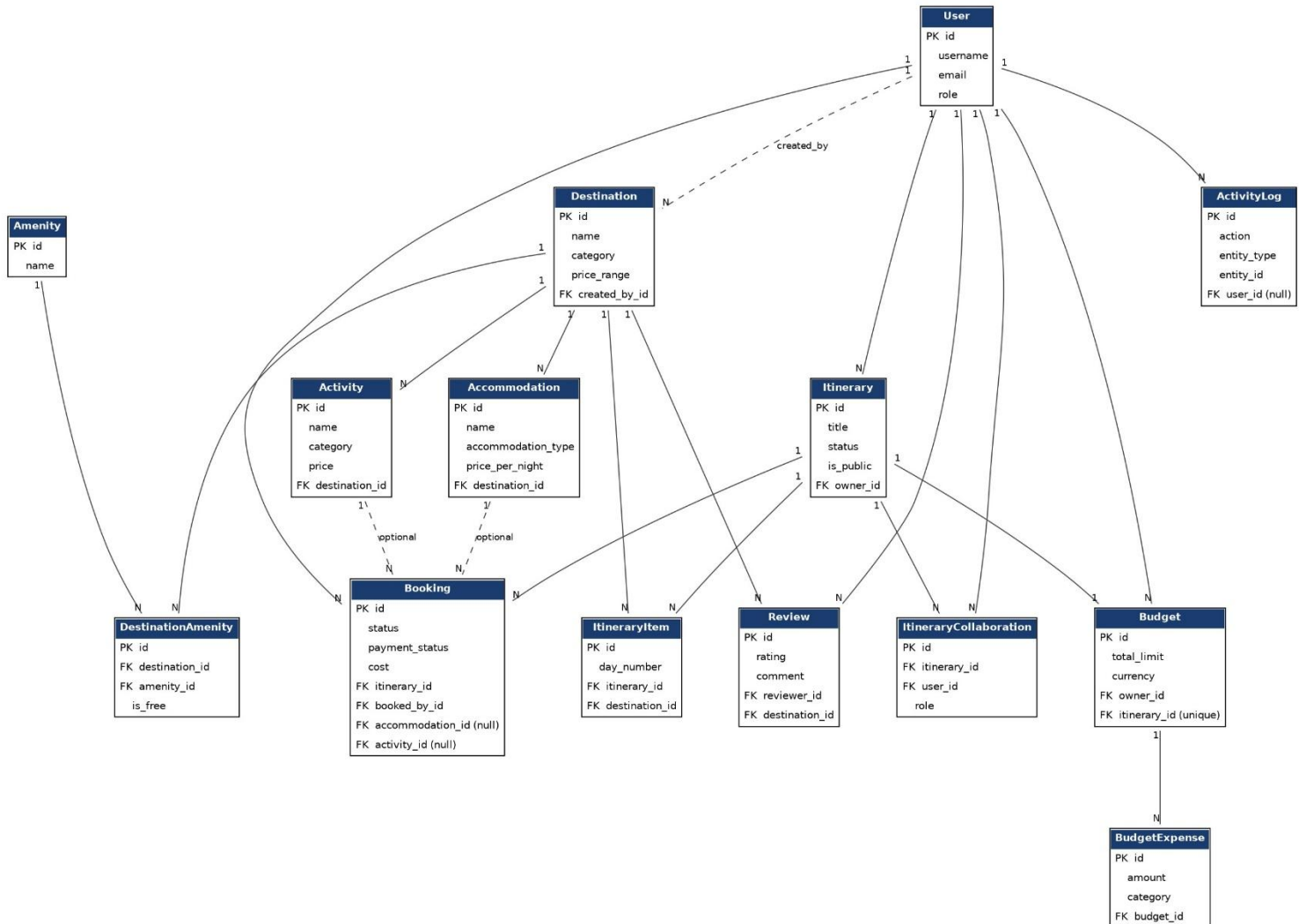


Planning Document

Travel Itinerary Planning & Booking API

1. ERD (Entity Relationship Diagram)



2. API Endpoint List

All endpoints are prefixed /api/v1/.

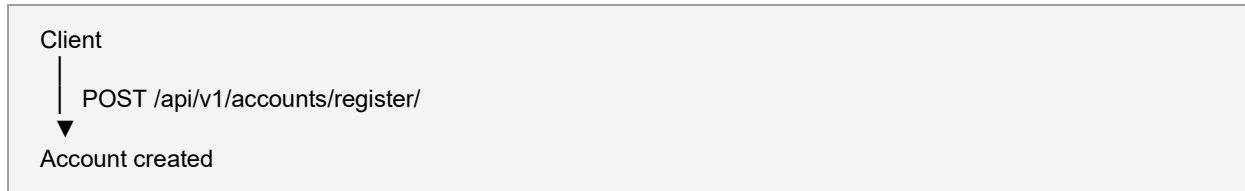
App	Method	Endpoint	Purpose
Accounts	POST	/api/v1/accounts/register/	Register a new account
	POST	/api/v1/accounts/login/	Login and obtain JWT tokens
	POST	/api/v1/accounts/login/refresh/	Obtain a new access token
	GET/PUT/PATCH	/api/v1/accounts/profile/	View or update your profile
	POST	/api/v1/accounts/password/change/	Change password

App	Method	Endpoint	Purpose
	POST	/api/v1/accounts/password/reset/	Request password reset
	POST	/api/v1/accounts/password/reset/confirm/	Confirm password reset
	GET	/api/v1/accounts/users/	List users
	GET/PUT/PATCH/DELETE	/api/v1/accounts/users/<id>/	Manage a specific user
Destinations	GET/POST	/api/v1/destinations/	List or create destinations
	GET/PUT/PATCH/DELETE	/api/v1/destinations/<id>/	Manage a destination
	GET	/api/v1/destinations/top_rated/	Get top-rated destinations
Itineraries	GET/POST	/api/v1/itineraries/	List or create itineraries
	GET/PUT/PATCH/DELETE	/api/v1/itineraries/<id>/	Manage an itinerary
	GET	/api/v1/itineraries/public/	List public itineraries
	GET/PATCH	/api/v1/itineraries/<id>/status/	View or update itinerary status
	GET/POST	/api/v1/itineraries/<id>/items/	Manage itinerary items
	GET/POST	/api/v1/itineraries/<id>/collaborators/	Manage itinerary collaborators
Bookings	GET/POST	/api/v1/bookings/	List or create bookings
	GET/PUT/PATCH/DELETE	/api/v1/bookings/<id>/	Manage a booking
	POST	/api/v1/bookings/<id>/confirm/	Confirm a booking
Reviews	GET/POST	/api/v1/reviews/	List or create reviews
	GET/PUT/PATCH/DELETE	/api/v1/reviews/<id>/	Manage a review
	GET	/api/v1/reviews/destination/<id>/	Get reviews for a destination
	GET	/api/v1/reviews/my_reviews/	Get the current user's reviews
Budgets	GET/POST	/api/v1/budgets/	List or create budgets
	GET/PUT/PATCH/DELETE	/api/v1/budgets/<id>/	Manage a budget
	GET	/api/v1/budgets/read-only/	View read-only budget data
	GET/POST	/api/v1/budgets/<id>/expenses/	List or create expenses
	GET	/api/v1/budgets/<id>/summary/	View budget summary

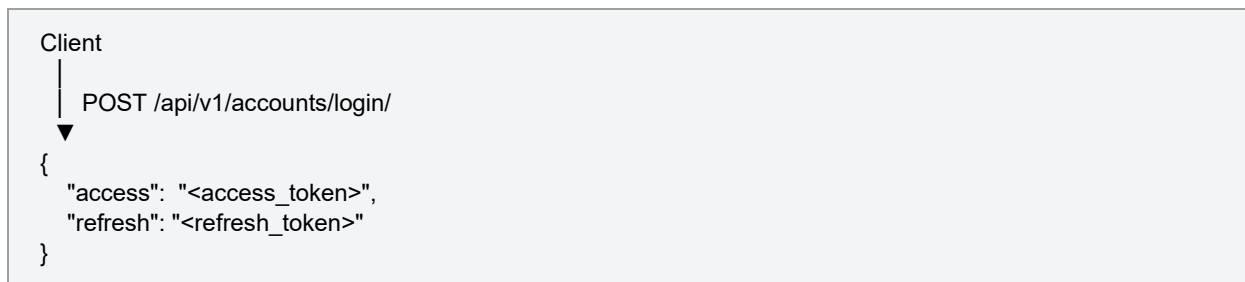
3. Authentication Flow

The API uses JWT (JSON Web Token) authentication with `django-rest-framework-simplejwt`.

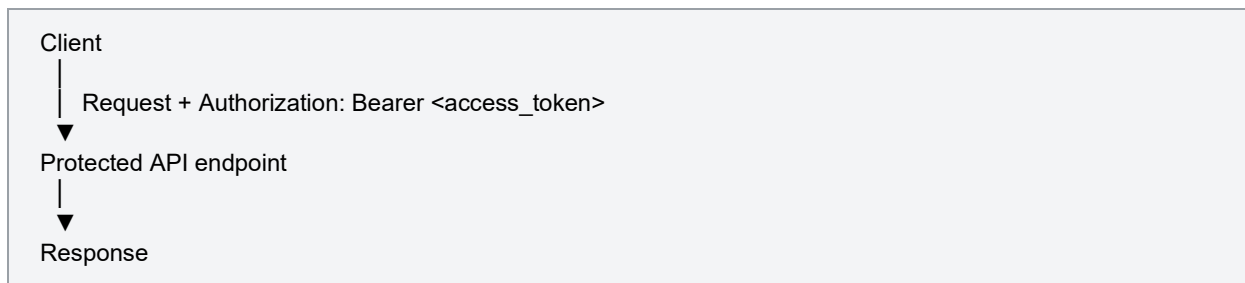
Step 1: Register



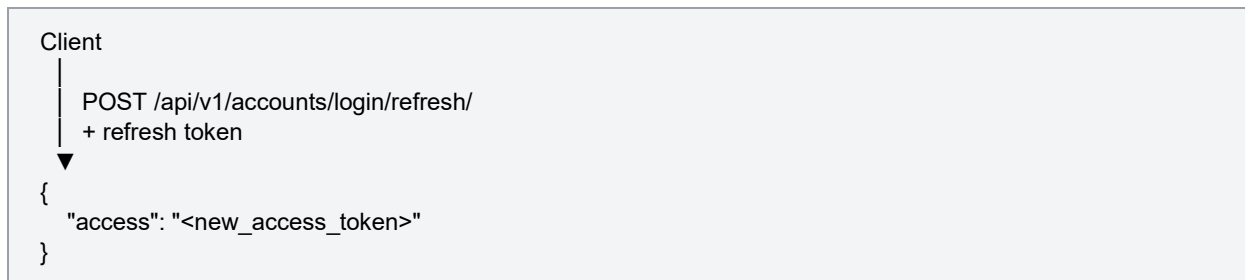
Step 2: Login



Step 3: Access a protected endpoint

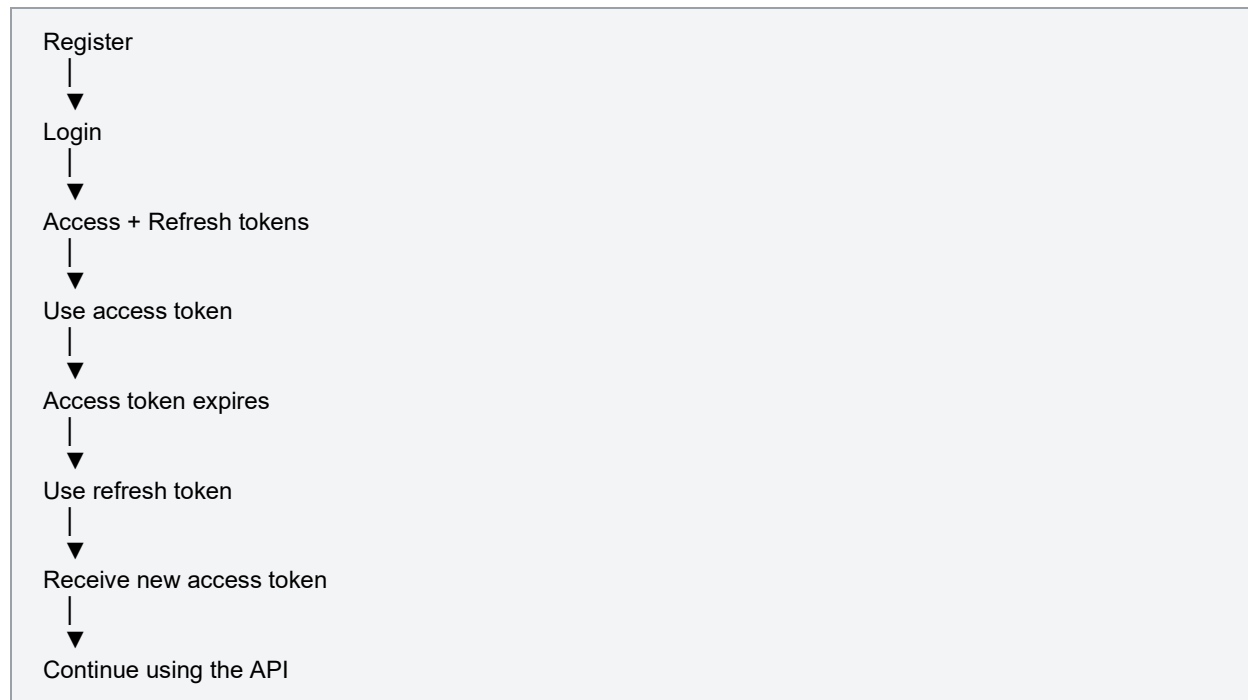


Step 4: Refresh an expired access token



The access token is used to authenticate requests to protected endpoints. When the access token expires, the refresh token is used to obtain a new access token. The refresh token is used only for this purpose.

Token lifecycle



4. Permission Matrix

Resource	Owner	Collaborator (editor/admin)	Collaborator (viewer)	Public	Staff/Admin
Itinerary	CRUD	Read/Write	Read only	Read (if public)	CRUD (all)
ItineraryItem	CRUD (own itinerary)	Write (per role)	Read only	—	CRUD (all)
Destination	Read only	—	—	Read	CRUD
Booking	CRUD (own)	—	—	—	CRUD (all)
Review	CRUD (own)	—	—	Read	CRUD (all)
Budget/Expense	CRUD (own)	—	—	—	CRUD (all)

5. URL Structure Organization

Each app owns its own `urls.py`, namespaced (`app_name`) and included under `/api/v1/<app>/` in the project root `urls.py`. ViewSets are registered with `DefaultRouter`; related sub-resources (itinerary items, budget expenses, collaborators) use nested paths under their parent's ID. Versioning (`v1/`) isolates the current API surface from future breaking changes.

6. Testing Strategy

Each app has its own `tests.py` using `APITestCase`, covering: model-level validation and constraints; serializer validation (including file size/type checks); endpoint behavior (CRUD, filtering, search, ordering, pagination) per role (owner, collaborator, public, staff); and custom permission classes tested directly against allowed/denied cases. 254 tests currently pass with 94% coverage.